

The Devil is in the Spurious Correlations: Boosting Moment Retrieval with Dynamic Learning

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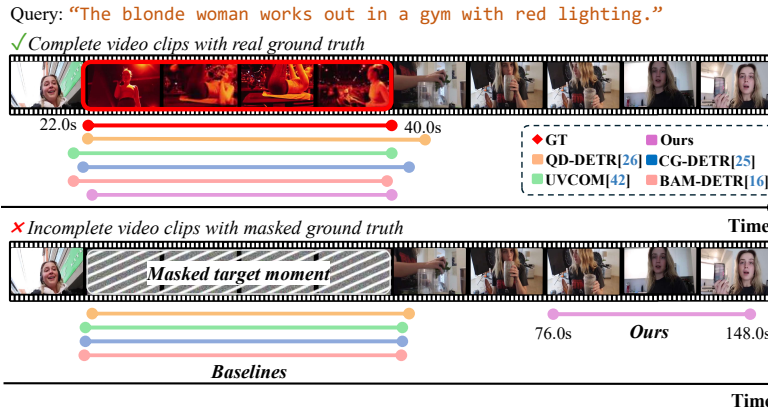


Figure 1. (a)

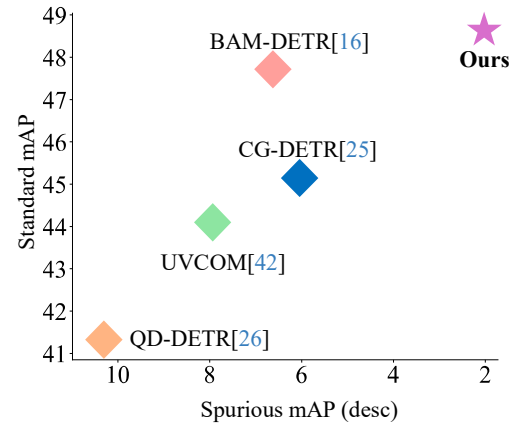


Figure 1. (b)

Figure 1. (a) Comparison of moment retrieval models under normal and spurious correlation videos by masking the content of target clips of video. We found the existing works are suffering from a crucial reason stems from the spurious correlation between the text queries and the moment context. Baselines predict the Spurious GT even if the target moments are *masked*. In contrast, *TD-DETR* predicts the segment near the mask with lower confidence. (b) To verify the issue of spurious correlation, we introduce the Spurious mAP as the metric. Our model achieves the best ratio of mAP to Spurious mAP.

Abstract

Given a textual query along with a corresponding video, the objective of moment retrieval aims to localize the moments relevant to the query within the video. While commendable results have been demonstrated by existing transformer-based approaches, predicting the accurate temporal span of the target moment is still a major challenge. This paper reveals that a crucial reason stems from the spurious correlation between the text query and the moment context. Namely, the model makes predictions by overly associating queries with background frames rather than distinguishing target moments. To address this issue, we propose a dynamic learning approach for moment retrieval, where two strategies are designed to mitigate the spurious correlation. First, we introduce a novel video synthesis approach to con-

struct a dynamic context for the queried moment, enabling the model to attend to the target moment of the corresponding query across dynamic backgrounds. Second, to alleviate the over-association with backgrounds, we enhance representations temporally by incorporating text-dynamics interaction, which encourages the model to align text with target moments through complementary dynamic representations. With the proposed method, our model significantly alleviates the spurious correlation issue in moment retrieval and establishes new state-of-the-art performance on two popular benchmarks, i.e., *QVHighlights* and *Charades-STA*. In addition, detailed ablation studies and evaluations across different architectures demonstrate the generalization and effectiveness of the proposed strategies. Our code is available at <https://github.com/xyangzhou/TD-DETR>.

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1. Introduction

Videos are a ubiquitous media format, but it can be time-consuming to browse through videos to localize specific moments. Using text to retrieve corresponding moments within a lengthy video[2, 54] plays an important role for entertainment, search, *etc.*

Existing moment retrieval approaches [15, 17, 19, 30, 49] leverage the DETR [4] detection transformer architecture to fuse text query into video representation. They show remarkable results in detecting the target moment by text-video alignment, but accurately predicting the temporal span remains a significant challenge. The learned video representations [17, 19] are aligned with the semantics of the text, thus generally biased to subjects or spatial objects. Consequently, the semantic elements in background frames are overly associated with textual queries. Such inappropriate association between irrelevant contexts and labels is the manifestation of *Spurious correlation* [1, 43], which are under-explored in moment retrieval tasks.

This work identifies such *spurious correlations* between the text queries and moment visual context as the root cause for this performance gap in moment retrieval. Specifically, learned SOTA moment retrievers [17, 19, 30] tend to associate text queries with the background frames rather than distinguishing the target moment. Figure 1 illustrates how such spurious issues lead to a sub-optimal performance for moment retrieval. For example, the SOTA method, BAM-DETR [17], predicts a similar temporal span for an ‘The blonde woman works out in a gym with red lighting.’ in both the original video and an altered version where the target moment is masked out. In the masked case, the segment near the mask is a more reasonable output but the baseline methods [17, 29, 30, 49] predict the masked segment from background frames with the learned spurious correlation, which is the most unrelated output to the text query.

To address the spurious issues, we propose a dynamic learning method with two novel strategies to mitigate spurious correlation. First, we propose a synthesis strategy that dynamically contextualizes the target moments for retrieval. We select similar video pairs and augment the samples by synthesizing target moments within similar contexts. This encourages the model to distinguish target moments from queries across more dynamic contexts, alleviating the over-association with background frames. Second, we propose to enhance video-text representations to reduce spurious dependencies on background frames by aligning text queries with temporal dynamics. To achieve this, we design a simple yet efficient temporal tokenizer to extract dynamic representations. Additionally, a text-dynamics interaction module is introduced to align dynamics with text queries, enabling the model to establish a stand-up correlation with complementary dynamic representations while

mitigating background bias.

Experiments on two challenging benchmarks, *i.e.*, QVHighlights and Charades-STA, show that our method outperforms existing methods by clear margins. Besides, *spurious correlation* in moment retrieval is validated to be largely alleviated by spurious evaluation. The detailed ablation analyses demonstrate the effectiveness of the proposed strategies to resolve the spurious issues.

In conclusion, our contributions to this work are summarized as follows:

- To the best of our knowledge, we are the first to investigate the spurious correlation in moment retrieval.
- We propose a dynamic learning framework for moment retrieval that mitigates spurious correlations by dynamically contextualizing target moments through novel video synthesis and enhancing representations with text-dynamics alignment.
- The proposed method achieves SOTA performance across all benchmarks. Besides, our model also provides a strong interpretation of alleviating *spurious correlations*.

2. Related Work

2.1. Moment Retrieval

Moment Retrieval aims to predict a temporal span within a video corresponding to a given natural language query [54]. Early methods had two stages: first, sampling candidate moments as proposals, and then scoring the proposals to obtain the final predictions [2, 8, 9, 23, 24]. However, more recent approaches are one-shot [18, 37, 47, 50–52, 56, 59] to predict the temporal span directly. Simultaneously, hybrid frameworks that integrate transformers for feature learning and fusion with CNNs for prediction have been widely studied. SnAG [31] explored the impact of cross-modal fusion on the scalability of video grounding models and proposed an efficient training scheme.

Although many moment retrieval methods have achieved remarkable performance, the majority primarily focus on architectural design, such as exploring new networks [16, 21, 27, 50, 52], improving contrastive learning, enhancing video-text alignment[18, 26], or refining attention mechanisms[31]. Generally, these methods focus on how to fuse text and video representations to improve the temporal span prediction but the overall performance are limited by the inefficient alignment architecture.

2.2. Detection Transformers

In the past several years, the wide adoption of detection transformers (DETR) [4] in video moment retrieval has gained remarkable performance [14, 17, 19, 25, 30, 48, 49]. While introducing the DETR framework to the task of moment retrieval, it demonstrated the effectiveness on QVHighlights dataset [19]. Furthermore, DETR-based ap-

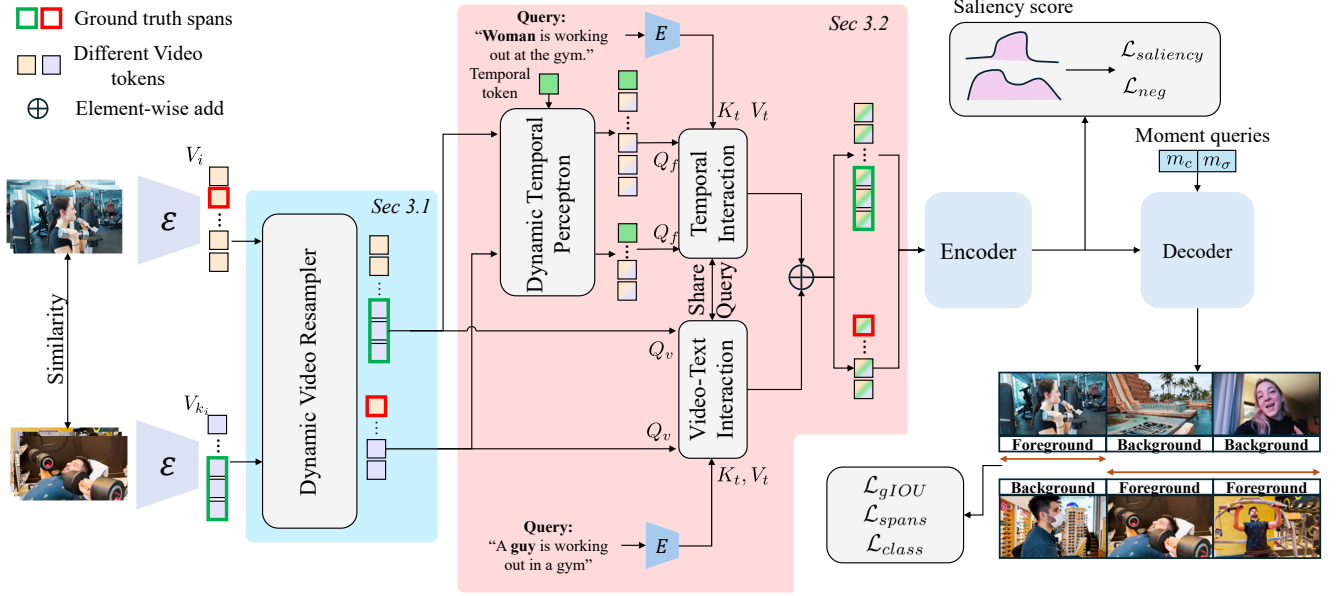


Figure 2. Overview of the proposed *TD-DETR*. Video pairs are sampled by their similarity and then forwarded into the Video Synthesizer to generate Dynamic Context (Section 3.1). Then, the Dynamic Context, *i.e.*, synthesized pairs, are enhanced by Temporal Dynamic Tokenization and interact with textual information (Section 3.2). Finally, the enhanced video tokens are sent to the transformer encoder-decoder with prediction heads to predict moments (Section 3.3).

proaches have explored the relationship between moment retrieval and highlight detection, achieving significant advancements in both tasks. However, they still struggle with inaccurate span predictions [33, 54], limiting their overall reliability. To address this issue, Zhang et al. [54] analyzes the distribution of target moments and highlights the problem of imbalance, while Otani et al. [33] investigate the skewed distribution of query representations and expose the overly query-dependent phenomenon of existing models. Unlike previous studies, we reveal that *spurious correlations* is the key cause of inaccurate span predictions, arising from the over-association between the text query and the background frames.

2.3. Spurious Correlation

One reason for the poor generalization of vision algorithms is that they are prone to memorizing patterns or contextual cues [3, 10, 11, 13, 39, 41]. These patterns or cues are often *spurious correlations* - misleading heuristics of the training data correlated with the majority of examples but does not hold in general [11]. While numerous approaches have been proposed to address this challenge [33, 42, 44, 45, 54, 55, 57, 58] in various domains, their impact on video understanding remains largely unexplored. For example, in image-based tasks, spurious correlations are often tied to spatial biases [1, 43], whereas in video-related tasks, such spatial biases are redundant in frames. This distinction is particularly crucial in retrieving moments from lengthy videos, where the goal is not merely

to recognize objects but to precisely localize temporal spans with text queries, making the study of spurious correlations in this context even more critical.

Despite in Moment Retrieval, Otani et al. [33] investigated the biases introduced by text queries and the neglect of semantic information, while Zhang et al. [54] explores the distribution of start and end moments. However, the issue of spurious correlation between the text and background frames is a key challenge of moment retrieval. In this work, we focus on this point and develop a new method by learning dynamics to alleviate this problem.

3. Methodology

The objective of moment retrieval is to localize a temporal span that semantically corresponds to a given text query. Consider a video represented by L vision tokens, denoted by $V = \{v_1, v_2, \dots, v_L\}$, along with a natural language description of W words, as $\{q_1, q_2, \dots, q_W\}$. A moment retrieval model predict a temporal span m with centre m_c and duration m_σ which is relevant to the textual query, and token-wise saliency scores $\{s_1, s_2, \dots, s_L\}$ of the text.

As shown in Figure 2, we propose a novel framework to learn *Temporal Dynamics* utilizing *DE*tectio*n TR*ansformer (**TD-DETR**). The video pairs are sampled by similarity and then forwarded into the video synthesizer to generate dynamic contexts. Then, the representations of synthesized pairs are enhanced through temporal dynamic interaction with texts. Finally, dynamic-enhanced representations are

fed into the transformer encoder-decoder with prediction heads to predict target moments.

3.1. Video Synthesizer for Dynamic Context

We introduce a novel video synthesizer to infuse the target moments with dynamic context for moment retrieval. Spurious correlations often stem from linking the moment’s context to the text query. We address the issue by synthesizing new samples for the target moments with more dynamic contextual variations.

Spurious Pair Selection. In order to synthesize a dynamic context aligned with the target moment, we construct a spurious pair of the target moment in a given video V . To ensure the challenge and rationality of the synthesized video, we select a video which is similar to V contextually.

In a training batch of N video samples, we sample the most similar video $\{V_k\}$ for every $\{V_i\}$ where $k \in [1, N], k \neq i$, for each $i \in [1, N]$ to construct a spurious video pair p_i . For any videos V_j and V_l from a batch, we employ cosine similarity to model the similarity relationship between videos as follows:

$$s_{j,l} = \frac{1}{L_j \cdot L_l} \sum_{p=0}^{L_j} \sum_{q=0, p \neq q}^{L_l} \frac{v_p^j \cdot v_q^l}{\|v_p^j\| \cdot \|v_q^l\|} \quad j, l \in [1, N]. \quad (1)$$

In Eq. 1, we denote v_p^j and v_q^l q -th clip of V_j and q -th clip of V_l , L_j and L_l for the length of V_j and V_l , $a_{j,l}$ for the average similarity between V_j and V_l . We select the most similar video V_k for every video V_i by $k = \arg \max_{k \in [0, N]} s_{[i, \setminus \{i\}]}$ in the batch.

Video Synthesis with Dynamic Context. After obtaining the spurious pairs, we synthesize a new video $\tilde{V}_i$ using the target moment m and the spurious pair p_i , where $p_i = \{V_i, V_k\}$. We then composite $\{V_i, V_k\}$ and dynamically refine the ground truth of the target moment m_i . Illustrated by Figure 3, given video V_i with L_i tokens and V_k with L_k tokens, on the one hand, we expect the sample from V_i and V_k with the completeness of target moment m_i of V_i . We first sample from V_i without ground truth tokens $NG \in [0, L_i] \setminus m_i$ with sampling ratio α , i.e., every token in NG has the same ratio α to be selected. We sample from the paired V_k with sampling ratio $1 - \alpha$, since V_k is irrelevant to ground truth GT_i .

Finally, we concatenate the two sets of sampled video tokens into $\tilde{V}_i$. On the other hand, we have the same process, except now we focus on V_k and GT_k with sampling ratio α and ratio $1 - \alpha$ to sample from the whole V_i and concatenate sampled tokens into $\tilde{V}_k$.

3.2. Dynamics Enhancement

The attention of DETR-like architecture associates text queries with visual representations. Unfortunately, this

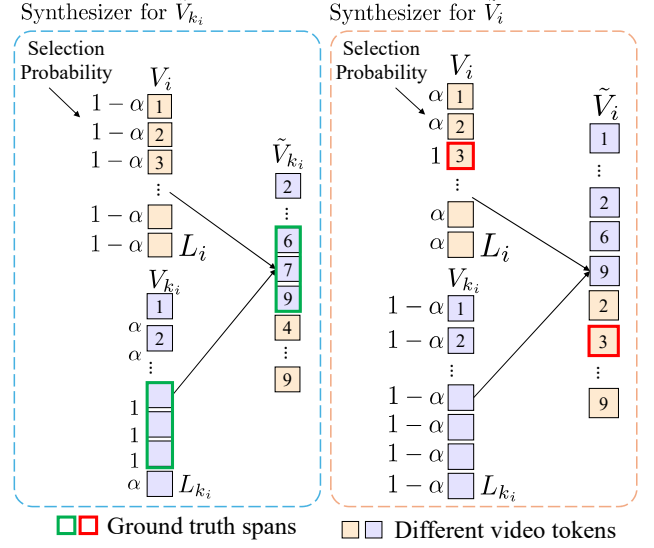


Figure 3. Illustration of Video Synthesis with Dynamic Context. The numbers in the boxes represent the token indices, indicating their sequential order. Vision tokens are selected based on their selection probabilities and concatenated while preserving the completeness of the ground truth spans, i.e., selection probabilities of ground truth spans are set to 1. The synthesized videos maintain their original length with a random bias.

tends to emphasize background frames, thereby exacerbating the *spurious correlation*. To this end, besides the dynamic context synthesis, we introduce a dynamics enhancement module to encourage our model to align text queries with temporal dynamic representations. Thus, the model considers not only the association of background frames but also background-independent dynamics. This alignment enables our model to establish a stand-up correlation between the query-related moment and its context.

Dynamic Tokenizer. To model the dynamic temporality, we introduce a simple yet effective strategy to tokenize the temporal dynamics, which is nearly cost-free to obtain temporal dynamic representations.

Given spurious pair $p_i = \{\tilde{V}_i, \tilde{V}_k\}$, the temporal dynamic tokenizer processes all video pairs in the same manner. Then, we use $\tilde{V}_i$ to explain the processing steps without loss of generality.

First, we concatenate a learnable token to the start of the video as follows:

$$\tilde{V}_i = \{st, \tilde{v}_1, \tilde{v}_2, \dots, \tilde{v}_{L_i}\}, \quad (2)$$

where st is a learnable token that captures the start signal of the video, preventing the loss of information from the first clip in the video representations. We then employ an element-wise difference to model the temporal dynamics.

$$T = \{\tilde{v}_1 - st, \tilde{v}_2 - \tilde{v}_1, \tilde{v}_3 - \tilde{v}_2, \dots, \tilde{v}_{L_i} - \tilde{v}_{L_i-1}\}, \quad (3)$$

where T is a learnable dynamic representation, it learns and focuses on the dynamic information in nearby video clips.

Text-Dynamics Interaction. After obtaining dynamic tokens T in Eq. 3 from Dynamic Tokenizer, we utilize cross-attention for both text-vision and text-dynamic representations with the same text query to mitigate the over-emphasis on video backgrounds.

In detail, the *query* for each cross-attention layer is prepared by linear projection of the video and temporal clips as follows:

$$Q_\delta = [p_\delta(\delta), \dots, p_\delta(\delta_{L_i})], \delta \in \{\tilde{V}_i, T\}. \quad (4)$$

The *key* and *value* are computed with the query text representation as $K_q = [p_k(q_1), \dots, p_k(q_W)]$ and $V_q = [p_v(q_1), \dots, p_v(q_L)]$, p_δ , p_k , p_v are linear projection layers for dynamic-enhanced *query*, temporal *query*, *key* and *value*. Then the cross-attention layer operates as follows:

$$\delta' = \text{softmax}\left(\frac{Q_\delta K_q^T}{\sqrt{d}}\right) V_q, \delta \in \{\tilde{V}_i, T\}, \delta' \in \{\tilde{V}_i', T'\}, \quad (5)$$

where d is the dimension of the projected *key*, *value* and *query*. To emphasize the learned temporal information, we utilize the weighted element-wise addition to inject text-guided temporal representation into text-guided dynamic-enhanced representation as follows:

$$\tilde{V}_i' = \beta \cdot \tilde{V}_i + (1 - \beta) \cdot T', \quad (6)$$

where β is a hyper-parameter to adapt the addition ratio between $\tilde{V}_i'$ and T' . Hence, from the spurious pair $p_i = \{\tilde{V}_i, \tilde{V}_k\}$, we obtain a new spurious pair $p_i' = \{\tilde{V}_i', \tilde{V}_k'\}$.

3.3. Network and Objectives

Our network structure (see Figure 2) follows previous work [19, 30]. It features the synthesis of dynamic contextual videos and the enhancement of dynamic representations.

Transformer encoder-decoder with prediction heads. Given spurious pair $p_i' = \{\tilde{V}_i', \tilde{V}_k'\}$, this module processes all video pairs in the same manner. Therefore, we use $\tilde{V}_i'$ to explain the processing steps without loss of generality. The encoder consists of T stacked typical transformer encoder layers, as in [19, 30], producing encoded representations E_{enc} . Our decoder, following [30], also uses T stacked typical transformer decoder layers, along with N learnable moment queries representing the centre m_c and duration m_σ . The decoder processes E_{enc} with the moment queries.

We adopt prediction heads following [30]. A linear layer predicts saliency scores from the encoded representations, and another linear layer handles negative pairs. From the decoder outputs, a 3-layer MLP with ReLU predicts the normalized moment center and duration, while a linear layer with softmax predicts the foreground.

Hungarian matching. Following [4, 19, 30], we perform Hungarian matching between the two predictions and two corresponding labels respectively. Give prediction $\hat{y}$ of $\tilde{V}_i'$ and corresponding to ground truth y , the optimal matching results between predictions and ground truths $\hat{\sigma}$ can be written as $\hat{\sigma} = \arg \min_{\sigma \in G_N} \sum_i^N C_{\text{match}}(y, \hat{y}_{\sigma(i)})$, where G is a permutation of predictions and ground truths pairs and C_{match} is the matching cost.

Loss Functions. We calculate the loss between $\hat{\sigma}$ and the ground truth y corresponding to $\tilde{V}_i'$. Following [19], the L_1 loss $\mathcal{L}_{L_1}$ and the gIoU [38] loss $\mathcal{L}_{gIoU}$ are used to measure the distance and overlapping between the predictions and a cross-entropy loss $\mathcal{L}_{cls}$ is used to measure classification. For highlight detection, we also use three loss functions which are margin ranking loss $\mathcal{L}_{\text{margin}}$, rank-aware contrastive loss $\mathcal{L}_{\text{margin}}$ and negative loss $\mathcal{L}_{\text{neg}}$.

The overall loss is shown as follows with λ_* as balancing coefficient:

$$\mathcal{L}_{\text{moment}} = \lambda_{L_1} \mathcal{L}_{L_1} + \lambda_{iou} \mathcal{L}_{gIoU} + \lambda_{cls} \mathcal{L}_{cls}, \quad (7)$$

$$\mathcal{L}_{hl} = \lambda_{\text{margin}} \mathcal{L}_{\text{margin}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}} + \lambda_{\text{neg}} \mathcal{L}_{\text{neg}}, \quad (8)$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{hl} + \mathcal{L}_{\text{moment}}. \quad (9)$$

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate our method on two widely-used benchmarks: QVHighlights [19] and Charades-STA [8], following the setup of prior works [19, 30]. *QVHighlights* has over 10,000 video-query pairs primarily from vlog and news content. We use the splits defined in Moment-DETR [19] and report results for both the *val* and *test* splits. *Charades-STA*, derived from the Charades dataset [40], has more than 18,000 video-sentence pairs. We follow the standard evaluation protocol from [8] and use two separate *training* and *testing* splits.

Standard Evaluation. We adopt the same metrics of Moment-DETR [17, 19, 30] for evaluation. Specifically, for Moment Retrieval, we report the mean average precision (mAP) at Intersection over Union (IoU) thresholds of 0.5 and 0.75, as well as the average mAP over IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05. We also include Recall@1 (R@1) at IoU thresholds of 0.5 and 0.75. For a fair comparison, we also include the *test* split of the QVHighlights dataset, which is evaluated on the CoDaLab competition platform [35]. Notably, performance at higher IoU thresholds, such as 0.7, serves as an indicator of more precise alignments between predicted moments and the ground truth.

Spurious Evaluation. To verify the spurious correlation with overly association the text with background, we replace the target moments in video content with random-valued masks while keeping the video duration unchanged,

Method	MR-R1		MR-mAP		
	@0.5	@0.7	@0.5	@0.75	Average
MCN [2] <i>ICCV'17</i>	11.41	2.72	24.94	8.22	10.67
CAL [6] <i>arXiv'19</i>	25.49	11.54	23.40	7.65	9.89
XML [18] <i>ECCV'20</i>	41.83	30.35	44.63	31.73	32.14
XML+ [19] <i>NIPS'21</i>	46.69	33.46	47.89	34.67	34.90
SnAG [†] [31] <i>CVPR'24</i>	59.79	48.10	58.63	44.37	42.71
<i>SnAG /w TD-DETR</i>	66.48	52.93	63.71	49.11	46.75
Moment-DETR [19] <i>NIPS'21</i>	52.89	33.02	54.82	29.40	30.73
UMT [25] <i>CVPR'22</i>	56.23	41.18	53.83	37.01	36.12
MomentDiff [20] <i>NIPS'23</i>	57.42	39.66	54.02	35.73	35.95
QD-DETR [30] <i>CVPR'23</i>	62.40	44.98	62.52	39.88	39.86
UniVTG [22] <i>ICCV'23</i>	58.86	40.86	57.60	35.59	35.47
CG-DETR [29] <i>arXiv'23</i>	65.40	48.40	64.50	42.80	42.90
UVCOM [46] <i>CVPR'24</i>	63.55	48.70	64.47	44.01	43.27
BAM-DETR [17] <i>ECCV'24</i>	64.53	48.64	64.57	46.33	45.36
<i>TD-DETR (Ours)</i>	64.53 _{±0.62}	50.37 _{±0.53}	66.21 _{±0.21}	47.32 _{±0.53}	46.69 _{±0.26}

[†]reproduced by the official code

Table 1. Performance comparison on QVHighlights *test* split. Our experimental results are averaged over three runs and ‘±’ denotes the standard deviation. For the compared methods, the results are copied from their original papers and we reproduce SnAG by the official code via slowfast+clip features.

Method	R1@0.5	R1@0.7
CAL [6]	44.90	24.37
2D TAN [56]	39.70	23.31
VSLNet [53]	47.31	30.19
IVG-DCL [32]	50.24	32.88
SnAG [†] [31]	65.72	37.32
<i>SnAG /w TD-DETR</i>	70.14	42.35
Moment-DETR [19]	53.63	31.37
Moment-Diff [20]	55.57	32.42
UMT [25]	48.31	29.25
QD-DETR [30]	57.31	32.55
CG-DETR [29]	58.40	36.30
BAM-DETR [17]	59.95	39.38
<i>TD-DETR (Ours)</i>	60.89	40.35

[†]reproduced by the official code

Table 2. Performance comparison on Charades-STA *test* split. Our experimental results are averaged over three runs. For all the compared methods, the results are taken from their original papers.

making the masked region the most unrelated moments to text queries. We then assess the model’s performance under these conditions and report the spurious metric of *Spurious R@1* and *Spurious mAP*, analogous to the standard R@1 and mAP. In this case, lower values are better, as they indicate less spurious correlation.

Implementation Details. Our method is implemented in PyTorch [34]. We follow the implementation of QD-DETR [30]. For all datasets, we use video features both extracted from SlowFast [7] pre-trained on Kinetics [5] and

Method	Spurious R1 ↓		Spurious mAP ↓		Standard mAP ↑	
	@0.7	@0.9	@0.75	Avg.	@0.75	Avg.
QD-DETR	9.35	5.29	9.90	10.40	41.82	41.22
Ours w/ QD	8.26	3.68	7.46	8.15	49.86	49.05
CG-DETR	4.65	1.29	5.55	6.14	45.70	44.90
Ours w/ CG	2.58	0.39	3.38	4.41	49.16	48.38
BAM-DETR	7.16	1.87	6.30	6.72	48.56	47.61
Ours w/ BAM	1.61	0.52	1.73	1.98	49.62	48.67

Table 3. Performance comparison on QVHighlights *val* split with random masks. The target clips in the video content are replaced with random-valued masks while keeping the video duration unchanged. We compare the baseline models—QD-DETR, CG-DETR, and BAM-DETR—against their respective versions integrated with our proposed *TD-DETR*, marked *Ours w/*, under the same experimental settings. The metrics here have been explained in section 4.1.

pre-trained CLIP [36] vision encoder, and text feature extracted from pre-trained CLIP [36] text encoder, following Moment-DETR.

- For *QVHighlights* dataset, we set batch size to 32 and an initial learning rate of $1e-4$ with weight decay of $1e-4$. We set the hidden size $d = 256$, layers of encoder/decoder $T = 3$, and moment queries $N = 10$. The model is trained for 200 epochs.
- For *Charades-STA*, we set batch size to 32 and use an initial learning rate of $1e-4$ with weight decay of $1e-4$. We set the hidden size $d = 256$, layers of encoder/decoder $T = 3$, and moment queries $N = 10$. The model is trained for 100 epochs and the learning rate is decayed to

$\frac{1}{10}$ every 40 epochs.

We use mini-batch and AdamW [28] gradient descent algorithm to optimize the network parameters, initialize weights with Xavier init [12], use dropout of 0.1 for transformer layers and 0.5 for input projection layers and set cost coefficient as the same as loss balancing coefficient on all datasets. For Dynamic Video Synthesizer, the selection Probability α is set to 0.5 and for Dynamic Temporal Identifier the β is set to 0.7 on all datasets. The total training time is approximately 8 hours on *QVHighlight* and 6 hours for *Charades-STA*, with a single NVIDIA RTX 2080 GPU.

4.2. Results on Standard Evaluation

As illustrated in Table 1 and Table 2, we compare *TD-DETR* with several baseline methods on *QVHighlights* and *Charades-STA* benchmarks.

Results on QVHighlights. On *QVHighlights* test split, we compare our model against proposal-based, proposal-free, and DETR-based methods, as shown in Table 1. As observed, our *TD-DETR* achieves state-of-the-art performance across all evaluation metrics. Notably, *TD-DETR* shows remarkable improvements compared to the other methods. Performance at higher IoU thresholds (*e.g.*, 0.7) provides a clearer indication of the alignment between predicted moments and the ground truth. Our *TD-DETR* outperforms the previous state-of-the-art model by a substantial margin, with improvements of up to 3.56% in $R@1@0.7$ and 2.14% in $mAP@0.75$.

Results on Charades-STA. We evaluate our method methods on *Charades-STA* test split, compared with proposal-based, proposal-free and DETR-based methods. As shown in Table 2, *TD-DETR* achieves state-of-the-art performance across all evaluation metrics. Notably, our method achieves improvements of up to 2.46% in $R1@0.7$ and 1.57% in $R1@0.5$ on this challenging benchmark. Recall that performance at high IoU thresholds stands for the precision with which predictions match the ground truth data.

4.3. Results on Spurious Correlation Evaluation

Moment retrieval models are often prone to inaccurate predictions due to spurious correlations, a typical example being when predictions rely overly on background cues. To investigate this issue, we adopt the spurious evaluation setting pre-defined in Sec 4.1.

We evaluate our methods on several strong baselines, including QD-DETR [30], CG-DETR [29] and BAM-DETR [17]. The results in Table 3 demonstrate the alleviation of spurious correlation issue, as indicated by the improved *Spurious R1* and *Spurious mAP* across all compared baselines while boosting the standard mAP . For instance, our methods improve QD-DETR [30] in *Spurious R1@0.9* from 10.40 to 8.15 (30.43%), with standard $mAP@0.75$ increasing from 41.22 to 49.05 (18.99%). While for BAM-

DETR [17], it achieves the most substantial improvement, improving *Spurious R1* to 1.61 and *Spurious mAP* to 1.73. The results on spurious correlation evaluation demonstrate the effectiveness of our proposed approaches to alleviate the spurious correlation issue and also the generalization across different baselines.

4.4. Qualitative Analysis

In this subsection, we analyze the impact of mitigating spurious correlations in our *TD-DETR* model. As illustrated in Figure 4, the entire video depicts a man speaking to the camera, while the target moment is described as: “A man wearing a cap backwards talking while some other videos appear in the bottom left corner of the screen.” Notably, baseline models fail to distinguish between “The man is talking” and “The man is talking while other videos appear in the corner,” indicating that background elements are overly correlated with the textual query. In contrast, our *TD-DETR* accurately predicts the target moment without being misled by background frames, demonstrating its ability to associate the information between text and the proper frames rather than relying on spurious correlations.

4.5. Ablation Studies

We conduct ablation studies on both *QVHighlights* and *Charades-STA* benchmarks to validate the effectiveness of each proposed component. *VSDC* and *TDEM* refer to Video Synthesizer for Dynamic Context and Temporal Dynamics Enhancement Module, respectively. As illustrated in Table 4, rows (b) to (c) show the effectiveness of each component compared to the baseline model (a), and (d) demonstrate the overall effectiveness of all the components. In detail, on *QVHighlight*, Video Synthesizer for Dynamic Context contributes improvement of 5.60% in $R1@0.7$, 9.18% in $mAP@0.75$ and 9.05% in average mAP while Temporal Dynamics Enhancement Module contributes improvement of 3.16% in $R1@0.7$, 4.19% in $mAP@0.75$ and 5.41% in average mAP for Moment Retrieval. With all components integrated, we observe a substantial 14.75%, 14.20% and 15.30% improvement in the $R1@0.7$, $mAP@0.75$ and average mAP , respectively. Moreover, the spurious correlation is also improved, where 30.43% in *Spurious R1@0.9* and 21.63% in *Spurious mAP*. The same phenomenon can be observed in *Charades-STA* as well. More experiments and analyses are included in the supplementary materials.

4.6. Generalization across Different Architectures.

We validate the generalization by incorporating our proposed approaches into different architectures. SnAG [31] is a hybrid framework, which integrates transformer for feature fusion and CNN for prediction, while QD-DETR [30] and BAM-DETR [17] follow DETR-like architecture.

As illustrated in Table 5, for the hybrid architecture, our

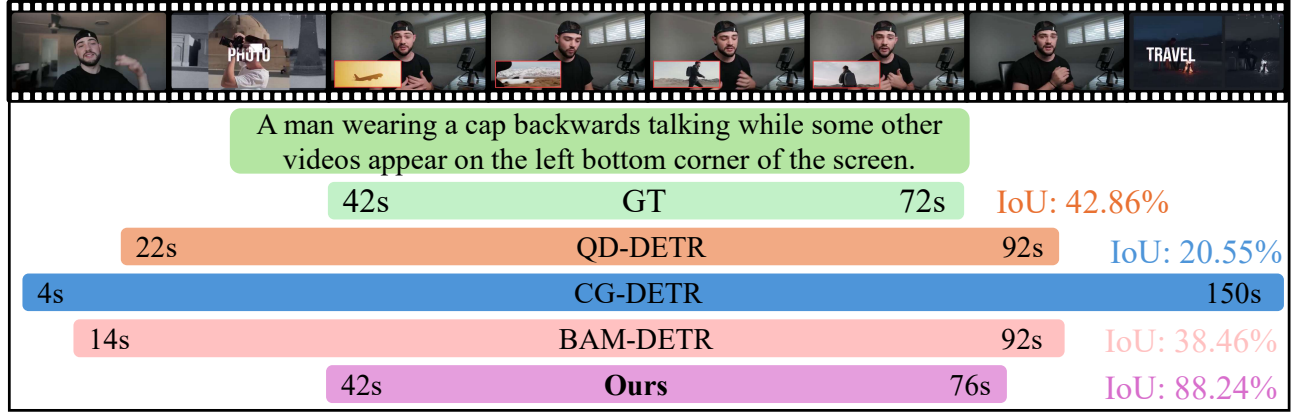


Figure 4. Visualization of results comparison between the baseline models and our *TD-DETR*. We display the relative position of ground truth and prediction moments in colourful boxes, where the green with GT stands for ground truth moment, the orange box with QD-DETR stands for our baseline predictions and the purple box with ours stands for our *TD-DETR*. To quantify the quality of predictions, we mark the start and end timestamps in the box and mark the IoU with the same colour as the models. Best read in colour.

	VSDC	TDEM	QVHighlight								Charades-STA				
			R1↑		mAP↑			Spurious R1↓		Spurious mAP↓		R1↑		Spurious R1↓	
			@0.5	@0.7	@0.5	@0.75	Avg.	@0.7	@0.9	@0.75	Avg.	@0.5	@0.7	@0.7	@0.9
(a)			61.12	46.77	62.45	43.66	42.54	9.35	5.29	9.90	10.40	57.31	32.55	25.72	6.31
(b)	✓		63.47	49.39	64.82	47.67	46.39	8.77	3.87	8.64	8.91	39.12	63.67	23.15	5.42
(c)		✓	62.93	48.25	64.22	45.49	44.84	8.84	4.0	9.10	9.56	38.51	60.80	24.03	5.73
(d)	✓	✓	65.88	53.67	66.43	49.86	49.05	8.26	3.68	7.46	8.15	60.89	40.35	22.13	4.82

Table 4. Ablation studies on QVHighlights *val* split and Charades-STA *test* split. *VSDC* and *TDEM* stand for Video Synthesizer for Dynamic Context and Temporal Dynamics Enhancement Module respectively. All the ablation results are averaged over three runs. We also reveal spurious metrics of the proposed modules.

Method	QVHighlights val			Charades-STA test	
	R1@0.7	mAP@0.75	mAP	R1@0.5	R1@0.7
CG	52.10	45.70	44.90	58.40	36.30
Ours w/ CG	53.25 $+1.15$	49.16 $+3.46$	48.38 $+3.48$	59.35 $+0.95$	37.84 $+1.54$
BAM	51.61	48.56	47.61	59.95	39.38
Ours w/BAM	52.87 $+1.26$	49.62 $+1.06$	48.82 $+1.21$	60.92 $+0.97$	40.25 $+0.87$
QD	46.66	41.82	41.22	57.31	32.55
Ours w/ QD	53.67 $+7.01$	49.86 $+8.04$	49.00 $+7.78$	60.89 $+3.58$	40.35 $+7.80$
SnAG	48.10	44.37	42.71	65.72	37.32
Ours w/ SnAG	52.93 $+4.83$	49.11 $+4.74$	46.75 $+4.04$	70.14 $+4.42$	42.35 $+5.03$

Table 5. Generalization across different architectures on both QVHighlight *val* split and Charades-STA *test* split. CG stands for CG-DETR, while QD stands for QD-DETR and BAM stands for BAM-DETR.

method via SnAG achieves improvements of up to 10.04% in R1@0.7 and 9.46% in mAP on QVHighlights and 6.73% in R1@0.5 and 13.48% in R1@0.7 on Charades-STA. For DETR-like architecture, our approach enhances QD-DETR with significant gains of 19.23% in R1@0.7 and 18.87% in mAP on QVHighlights and 6.25% in R1@0.5 and 23.96% in R1@0.7 on Charades-STA. Our strategies significantly improve the performance of both hybrid and DETR-like baselines on both benchmarks, demonstrating the generalization of our approach.

5. Conclusion

Although existing transformer-based approaches have demonstrated remarkable performance, they still struggle with spurious correlation, which overly associates the textual query with the background frames rather than the target moment. To address this issue in moment retrieval, we introduce a novel dynamic learning method with two strategies. First, we propose a novel video synthesis strategy that constructs a dynamic context for the relevant moment. This synthesis strategy enables our model to attend to the target moment corresponding to the query across various dynamic video contexts. Second, we enhance the representation by learning temporal dynamics aligned with texts. In addition to visual features, text queries are aligned with dynamic representations, encouraging our model to establish a non-spurious correlation between the query-related moment and its context. Extensive experiments and detailed ablation studies on both the QVHighlights and Charades-STA benchmarks validate the effectiveness and generalization of the proposed *TD-DETR*, demonstrating significant alleviation of spurious correlation and superior performance in moment retrieval.

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